

Improving Surface Enhanced Raman Spectroscopy (SERS) Preprocessing and Analysis

André Dias¹, Diogo Cachetas^{2,3,4}, Laura Rodriguez-Lorenzo², and Miguel Rocha^{3,4}

¹ University of Minho, Braga, Portugal

² International Iberian Nanotechnology Laboratory (INL), Av. Mestre José Veiga, Braga, Portugal

³ Centre of Biological Engineering, University of Minho, Braga, Portugal

⁴ LABBELS – Associate Laboratory, Braga Guimarães, Portugal

Abstract. Antibiotic residues, particularly sulfonamides, are recognized as emerging environmental contaminants due to their widespread usage in human and veterinary medicine, and their persistence in aquatic environments. Traditional detection techniques for antibiotic residues are often limited by high costs, complex procedures, and lack of field applicability. Raman spectroscopy, particularly Surface-Enhanced Raman Spectroscopy (SERS), offers a promising alternative, providing rapid, non-destructive, and sensitive molecular analysis. This project aims to establish a robust preprocessing pipeline for Raman spectra, optimizing signal quality and enabling downstream analysis using machine learning. Ultimately, the integration of SERS and advanced computational tools facilitates the detection of antibiotic residues in water matrices and sets the stage for the development of intelligent, automated systems in environmental monitoring.

Keywords: Preprocessing · Water · Sulphonamides · Surface-Enhanced Raman Spectroscopy (SERS).

1 Introduction

Antibiotic residues have been listed as contaminants of emerging concern, being disseminated through multiple different sources such as hospitals, municipal sewage and, animal husbandry they easily disperse and affect many environments [1]. The One Health concept states that health, in general, is affected by many factors that are intertwined and largely interdependent, specifically, humans, animals, plants, and the environment, placing water in the central role as a common link between these compartments [2]. Most antibiotic molecules are not fully metabolized by humans or animals, they are secreted from urban hubs like hospitals, aquaculture facilities, wastewater treatment plants, livestock farms, and many others where antibiotics are used daily, these enter wastewaters and create antibiotic residues hotspots in the latter receiving environments [3–5]. This is aggravated by the fact that some antibiotics are poorly biodegradable and can

persist in the environment even at trace levels [6]. Sulfonamides belong to a class of synthetic antibiotics that have a varied range of biological activities, these are generally stable and easy to prepare, making them a valuable asset in medicine for treating human and animal diseases [7, 8]. A review by Maghsodian et al. (2022) [9] showed insights into the presence of antibiotics in water, sediments, and organisms in aquatic environments. Antibiotics were found at a concentration of <1 ng/L– 100 μ g/L in aqueous media, with sulphonamides being the most abundant in rivers worldwide (30%; 191.11 ng/L average concentration) [9]. Consequently, these have also been referred to in the World Health Organization's (WHO) sixth revision of the "Critically Important Antimicrobials for Human Medicine" list [10]. Therefore, sulphonamides may pose an unprecedented group of antibiotics to monitor antibiotic residue contamination.

A wide range of techniques for antibiotic detection have been used in the past years, capable of detection with high precision and ease, namely liquid chromatography-based, mass-spectrometry, ultraviolet-visible spectroscopy, optical immunoassay methods, resonance Rayleigh scattering, and optical fibre-based methods[11–13] However, most of them face many limitations, such as costly equipment, require specialized training, extensive pretreatment, and are unable to perform on-field [12].

Raman spectroscopy has seen an increase in use and has become one of the most utilized optical spectroscopic techniques. Not only is it fast and non-destructive but it generally requires simpler sample preparation. It provides detailed information about chemical structure, phase and polymorphy, crystallinity and molecular interactions [14–16]. Raman spectroscopy is a technique that utilizes inelastic scattering, a phenomenon in which photons from an incident monochromatic light with certain frequency interact with a molecule and a very small fraction of this light is scattered, resulting in an frequency shift (to higher or lower frequencies) that provides molecular vibrational information. However, traditional Raman spectroscopy suffers from inherent weak signal intensity and noise interference [17–22]. Surface-enhanced Raman spectroscopy (SERS) overcomes these limitations thanks to the enhancement of the Raman signal when molecules are close to plasmonic nanostructures made from mainly gold and silver. SERS is a powerful analytical technique widely used for detecting trace amounts of antibiotic residues, including sulphonamides. The analytical applicability of SERS is strongly connected with the development of very active SERS substrates to be able to achieve the single-molecule detection. Due to their unique morphology, gold nanostars (NSTs) effectively concentrate electromagnetic fields at their tips, leading to significant signal amplification and improved detection sensitivity, amplifying Raman signals by factors ranging from 10^5 to 10^{10} [20, 23, 24].

Despite the advantages described above, one of the biggest challenges in the implementation of SERS in environmental monitoring is the number of species that can be detected simultaneously. If the number of molecules, including targets and background species, is high, the overlapping of vibrational modes of different molecules is very likely together with background noise, cosmic ray in-

interference, and signal variability. In this situation, the interpretation of the vibrational spectrum becomes difficult or nearly impossible. Effective preprocessing and signal enhancement methods are essential for reliable data interpretation [17–22].

Thus, the end goal is to utilize the treated data for machine learning tasks towards more informative and efficient models since these have, in the last few years, shown auspicious results, when it comes to the analysis of complex, large, and high-dimensional data [25, 26].

2 Methods

2.1 Preprocessing

As stated in the above section, SERS spectra are prone to contain artefacts and redundant information that will negatively impact data analysis. Thus, to mitigate the impact of the aforementioned preprocessing methods are required to enhance the quality of the spectra. These methods are mainly applied to the background noise enabling its removal and minimizing its impact on the data analysis through the employment of statistical and computational techniques that are also capable of improving the signal-to-noise ratio (SNR) making the processed data more informative and comprehensive [27, 28]. There are many available methods for preprocessing Raman spectra, highlighting the need for an optimized pipeline with generalization ability for integrated analysis. For this, the next section will focus on the most well-known methods for Raman spectra preprocessing, namely, cropping, spike removal, denoising, baseline correction and normalization [29]. Additionally, machine learning-based methods are also emerging to enhance automation and robustness in preprocessing routines [29–31].

Cropping allows to remove redundant features and reduce data dimensionality by focusing on the “fingerprint” spectral region of interest, generally between 200 and 2000 cm^{-1} , as this range holds the most relevant molecular information. While not mandatory, selecting this region ensures that key spectral features are preserved [27, 32].

Spike removal, also known as cosmic ray removal, is a crucial step, as these artefacts can significantly adulterate the data. Methods for spike removal can be categorized into single-spectrum approaches, which filter spikes from an individual spectrum, and multi-spectrum approaches, which compare multiple spectra to detect and eliminate outliers, one of the most straightforward and standardized techniques is the Whittaker Hayes method, calculating the modified Z-scores to locate spikes and then removing them through the use of a moving average filter [18, 33–36].

Denoising algorithms further refine the spectra by improving signal-to-noise-ratio (SNR). Common methods include Savitzky-Golay smoothing [29, 37, 38], which applies polynomial least squares fitting within a moving window, although not originally used for denoising, autoencoders have also proved themselves as

an effective method for capturing non-linear relationships and enabling quality denoising while improving the Raman spectra [29, 22, 39–43].

Baseline correction is a critical step for accurate Raman analysis, as background interference from fluorescence and instrumental noise can influence spectral features. Techniques like polynomial fitting, asymmetric least squares (AsLS), and iterative smoothing-splines with root error adjustment (ISREA), which isolates molecular information and enhance peak identification [29, 44–46].

Normalization ensures consistency across spectra by accounting for variations in intensity levels. Single-spectrum methods, such as min-max or log min-max normalization, vector normalization, and peak normalization, standardize spectra without external references. Multi-spectrum techniques, including multiplicative scatter correction and extended multiplicative signal correction, use reference data for improved accuracy [29, 43].

Table 1. Raman Spectroscopy/SERS Preprocessing Steps and Methods

Methods	Objective	Methods/Techniques
Cropping	Reduce dimensionality and focus on spectral region	- Selection of the "fingerprint" region (200–2000 cm^{-1}).
Spike Removal	Eliminate cosmic rays spikes	- Single and multiple-spectrum approaches; - Whittaker-Hayes method (Z-score + moving average filter).
Denoising	Improve signal-to-noise ratio (SNR)	- Savitzky-Golay smoothing; - Autoencoders; - Center weighted moving average (CWMA).
Baseline Correction	Remove background interference	- Polynomial fitting; - Asymmetric Least Squares (AsLS); - ISREA method.
Normalization	Standardize intensity levels	- Min-max, log min-max, vector, peak normalization; - MSC; - EMSC.

2.2 Multivariate Data Analysis

Data interpretation and visualization are crucial to better understand the underlying patterns of the data while, simultaneously, allowing to iteratively fine-tune the preprocessing parameters. These methods may be used towards feature extraction and improving visualization and interpretability, by using intrinsic variable correlations. Different methods for multivariate analysis may be divided into two main tentative categories [29, 47, 48]

2.2.1 Unsupervised Machine Learning A machine learning method that is used to evaluate patterns and structures with no respect to data labels. Some relevant techniques include [26], dimensionality reduction methods such as, Principal Component Analysis (PCA) allowing to linearly project a reduced representation of the data capturing key spectral variations with principal components; Uniform Manifold Approximation and Projection (UMAP), maintains local and global data structure while being computationally efficient; and t-Distributed Stochastic Neighbour Embedding (t-SNE) which effectively preserves relationships in high-dimensional data but requires further parameter tuning [29, 49–54].

On the other hand, K-means clustering, is a simple algorithm that allow to search for clusters in the data in accordance to its proximity to each cluster centroid; Hierarchical Cluster Analysis (HCA), is a method that allows to group similar objects into clusters and forming a hierarchy represented in a dendrogram [29, 55].

2.2.2 Supervised Machine Learning Supervised methods, unlike unsupervised approaches, labelled data is required. There are mainly two distinct phases: (1) the training phase, to find the patterns in the data using a training data set followed by validation of the model for example with stratified k-fold cross validation ensuring that relative class frequencies are maintained across each series and fold validation, using stratified sampling, and (2) the prediction phase, where the learned patterns are fitted to the test set, usually for regression and classification tasks.

Both supervised and unsupervised analysis are commonly used together for computational efficiency and avoid overfitting [29, 56]. Conventional methods include, Support Vector Machines (SVMs), that allow to construct optimal decision boundaries in high-dimensional space to classify Raman spectra, making them effective for distinguishing between similar spectral profiles; Linear Discriminant Analysis (LDA) optimizes class separation by finding linear combinations of features that maximize inter-class variance while minimizing intra-class variance, as previously stated, although LDA is computationally efficient, it requires a larger number of observations than variables, often achieved using dimensionality reduction through aforementioned unsupervised methods [29, 57, 58].

2.3 Pipeline

The preprocessing pipeline is divided into six steps. First, the spectrum is cropped between 300 and 2000 cm^{-1} to contextualize the baseline correction algorithm. Next, cosmic ray removal is performed using the Whittaker-Hayes method, followed by Savitzky-Golay denoising to improve the signal-to-noise ratio. Baseline correction and subtraction are applied using DRPLS (Dynamic Reweighted Penalized Least Squares), then normalization is done based on the intensity of a reference peak in the Raman shift. Finally, a second cropping step narrows the range to 400–2000 cm^{-1} , isolating key molecular information.

3 Dataset

A dataset for classification tasks was used, with a total of 729 spectra from water matrices, divided into almost equal parts by five labels: Control, Mixture, Sulfamethoxazole, Sulfapyridine, and Sulfathiazole (Fig. 1). Control does not contain any antibiotics while, in contrast, the mixture label encompasses the latter three sulfonamide subgroups.

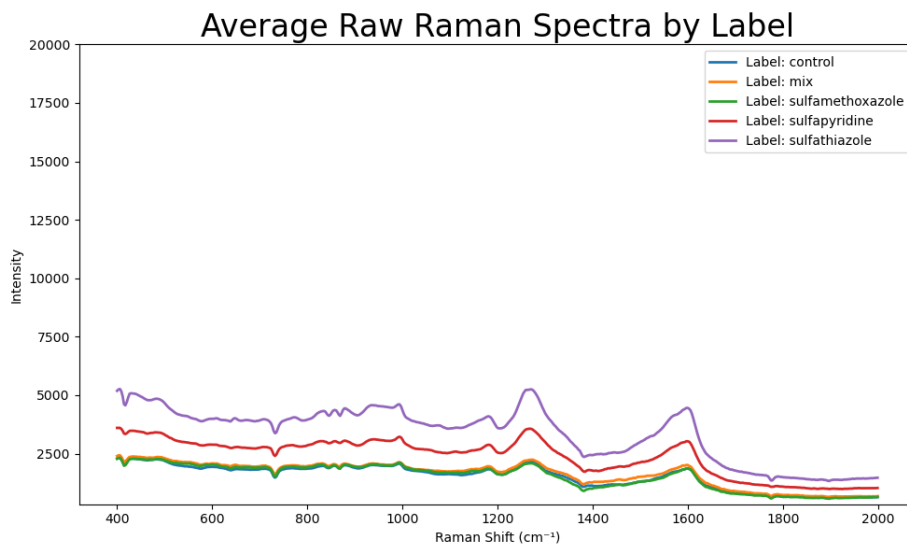


Fig. 1. Average Raw Spectra by Label

4 Results

After applying the preprocessing pipeline to the raw Raman spectra, we observed a significant improvement in data quality. Many of the artifacts and noise were eliminated, the baseline was corrected, and the true chemical signals became much more evident. The resulting spectra are more informative, with well-defined and comparable peaks across different samples, regardless of absolute signal intensity. This enhancement is essential to ensure the accuracy and reliability of both statistical analyses and machine learning algorithms. In summary, the preprocessing pipeline enhances the information contained in this preprocessed representation while mitigating the effect of artefact and external interferences, enabling precise and accurate information extraction, enabling the data for unsupervised and supervised learning methods (Fig. 2).

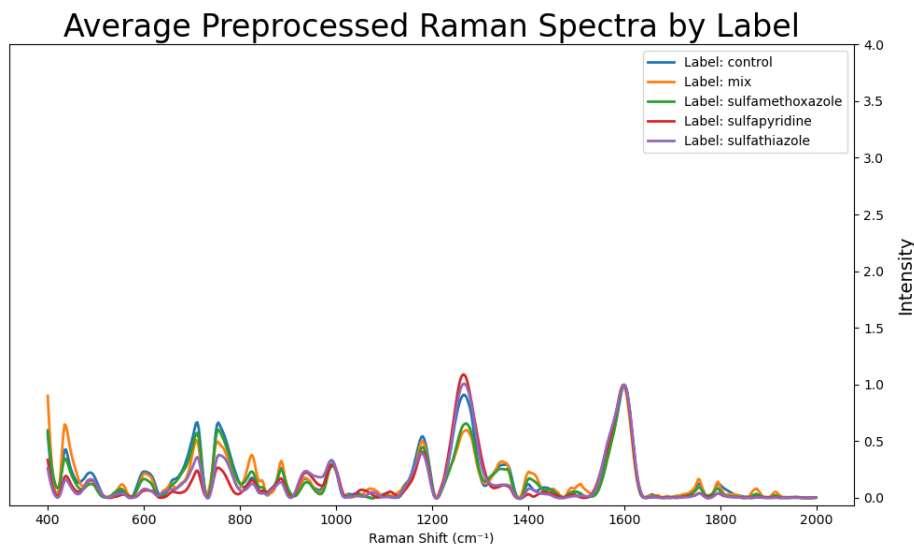


Fig. 2. Average Preprocessed Raman Spectra by Label

4.1 Dimensionality Reduction

As a means to validate and analyze the effectiveness of the preprocessing pipeline, in order to infer about the occurrence of clusters, dimensionality reduction techniques -PCA, t-SNE, and UMAP - were applied to both the raw data, the preprocessed Raman spectra, and to the preprocessed spectrum after feature selection with the Pearson correlation matrix.

4.1.1 Raw and Preprocessed Data Clear differences emerged between the two types of data (Fig. 3). In the case of the raw data, PCA revealed that the majority of the variance was explained by a single component (PC1: 99.2%, PC2: 0.3%), indicating that the dataset was highly redundant, with meaningful information likely masked by background noise. This outcome is expected, as the unprocessed data often contains strong but uninformative signals dominating the variance. In contrast, after the preprocessing pipeline was applied, the PCA variance distribution became more balanced (PC1: 43.8%; PC2: 31.2%), suggesting that the noise had been reduced and a more informative structure was recovered. The data now exhibited multiple meaningful directions of variation, rather than being dominated by a single trend. When examining the two nonlinear dimensionality reduction methods, t-SNE and UMAP, the most noticeable difference lies in the way the data clusters and organizes itself. In the raw data, no discernible pattern or structure could be

observed in the projections, reinforcing the idea that the data was dominated by noise. However, after preprocessing, both t-SNE and UMAP revealed clearer groupings and separations, indicating that the pipeline was effective in enhancing

the underlying signal and enabling the extraction of meaningful patterns. These results suggest that the preprocessing steps not only reduced redundancy and noise but also improved the data’s structure in ways that are detectable by both linear and nonlinear dimensionality reduction techniques.

4.1.2 Feature Selection By employing a Pearson Correlation Matrix without using any labels, this technique identifies strong relationships between variables and removes the most highly correlated wavenumbers, specifically those with a correlation above 0.95. This approach reduces dimensionality, helps prevent overfitting in machine learning models, and increases computational efficiency, although it may lead to slight decreases in predictive accuracy, as some informative features may be discarded in the process. This effect is reflected in the differences observed when applying dimensionality reduction techniques to both the original preprocessed dataset and the feature-selected version (Fig. 3). Notably, t-SNE and UMAP show less defined clustering after feature selection, suggesting a loss of structure. Additionally, PCA reveals that most of the variance is captured by a single component (PC1: 81.1%, PC2: 6.0%), indicating that the retained features are highly correlated.

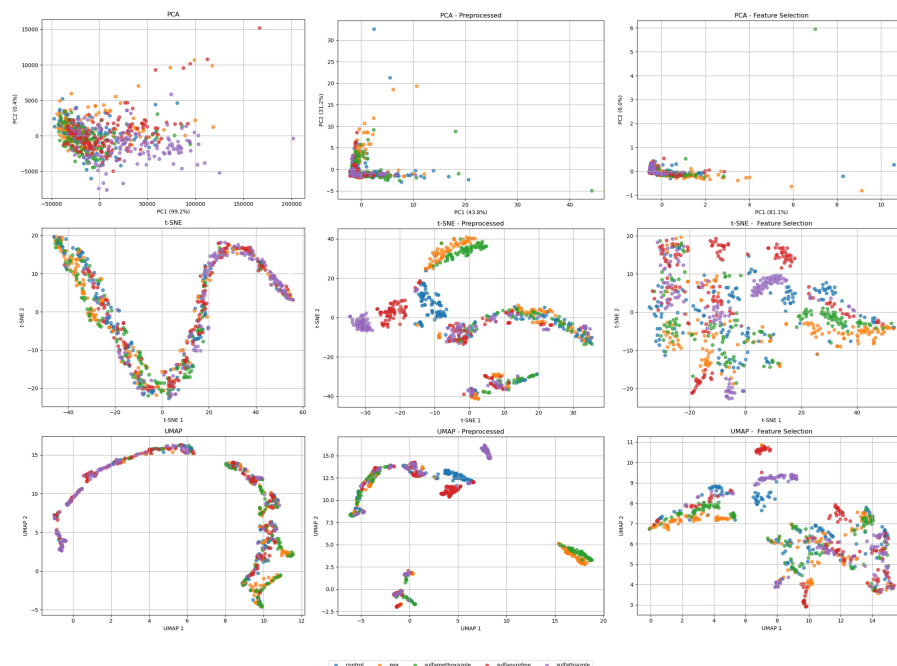


Fig. 3. Dimensionality Reduction on RAW, PreProcessed and Feature Selected datasets

4.2 Machine Learning

4.2.1 Exploratory To explore various machine learning models without any parameter tuning and determine which ones performed best on our dataset, primarily based on the accuracy metric, the LazyPredict library was used, with an 80%/20% train-test stratified split. As a next step, the top three models (RandomForestClassifier, LGBMClassifier, and ExtraTreesClassifier) were further evaluated, still without any parameter tuning, but this time using Stratified 5-Fold Cross Validation. This validation was performed on the training data from the initial split, leaving the test set untouched for the final deployment evaluation. The average values of common evaluation metrics, Accuracy, Precision, Recall, and F1 Score, were then calculated across the five folds. This process was applied to both the preprocessed dataset and the feature-selected dataset, allowing us to compare which setting yielded the best model performance. The model expected to deliver the best results was the ExtraTreesClassifier, and this was confirmed on the preprocessed dataset, where it achieved an accuracy of 80%. On the other hand, the feature-selected dataset yielded slightly better results with the LGBMClassifier (77% accuracy), while ExtraTrees achieved 76%. However, since the difference was minimal, we ultimately proceeded with hyperparameter tuning and deployment using the ExtraTreesClassifier.

4.2.2 Fine Tuning and Deployment To improve the performance of the chosen model, a grid search was conducted using the GridSearchCV function from the scikit-learn library. The search space included combinations of key hyper-parameters: the number of trees (`n_estimators`), maximum tree depth (`max_depth`), minimum number of samples to split a node (`min_samples_split`), minimum number of samples required at a leaf node (`min_samples_leaf`), and the strategy for selecting features (`max_features`). A stratified 5-fold cross-validation (StratifiedKFold) was applied to ensure that class proportions were preserved across folds. The training set was used exclusively during this process, and accuracy was chosen as the scoring metric. The grid search identified the best combination of hyperparameters based on the mean cross-validated accuracy score.

These optimal parameters were then used to evaluate whether any improvements could be achieved before the final deployment. Although small, this was verified, the preprocessed dataset went up to an accuracy of 81% while the feature selected increased to 77%.

Lastly, these parameters were used on the final model deployment, training it with the full training data and evaluated on the held-out 20% that were left aside, this evaluation was conducted separately on both the preprocessed dataset and the feature selected dataset, enabling a direct performance comparison (Fig. 4).

The preprocessed dataset model achieved an overall accuracy of 84%, with a macro-averaged F1-score also of 84%. As shown in the confusion matrix, the predictions were well balanced across all five classes. The best performance was seen in the control and mix classes, both achieving an F1-score of 0.89, while sulfamethoxazole and sulfathiazole had slightly lower scores of 78% and 82%,

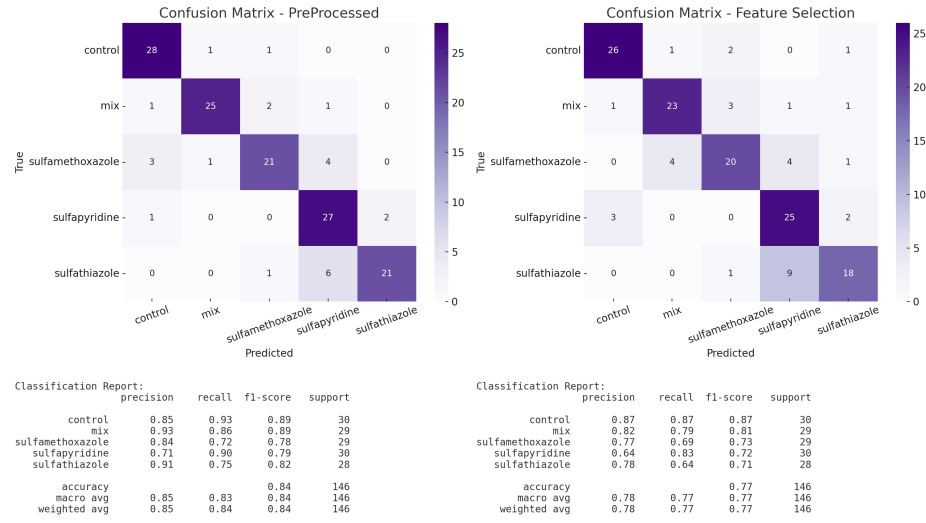


Fig. 4. Matrices de confusão e classification reports dos modelos antes e após seleção de features

respectively. The class sulfapyridine showed reasonable performance with an F1-score of 79%. These results suggest that the model was able to generalize well to unseen data, with strong precision and recall across all classes.

In contrast, the model trained on the feature-selected dataset achieved a lower overall accuracy of 77%, and a macro-averaged F1-score of 77%. The class-wise performance was more variable in this case. While control maintained a strong F1-score of 87%, sulfapyridine and sulfathiazole showed reduced performance, with F1-scores of 72% and 71%, respectively.

The confusion matrix shows more misclassifications in this scenario, particularly among the antibiotic-treated classes, suggesting that some useful information might have been lost during the feature selection step.

5 Conclusions

This study demonstrated a comprehensive pipeline for preprocessing, analyzing, and modeling Raman spectral data. The preprocessing played a critical role in improving the interpretability of the data, as evidenced by the transformation of variance distributions in PCA and the emergence of structured clusters in t-SNE and UMAP projections.

However, the application of feature selection based on Pearson correlation, while effective in reducing dimensionality, led to a noticeable degradation in the

structural representation of the data, particularly evident in the less defined clusters produced by t-SNE and UMAP.

Machine learning experiments further supported these findings. The Extra-TreesClassifier was selected for fine-tuning based on cross-validation results and underwent hyperparameter optimization using Grid Search. Evaluation on the held-out test set revealed that the model trained on the fully preprocessed dataset achieved superior performance compared to the one trained on the feature-selected version. Specifically, it attained an accuracy of 0.84 and a macro-averaged F1-score of 0.84, with well-balanced class-wise performance. In contrast, the model trained on the feature-selected data reached an accuracy of 0.77 and a macro F1-score of 0.77, with greater variability in class-specific metrics and a higher rate of miss classification.

In conclusion, while feature selection improved computational efficiency and prevented overfitting, it came at the cost of predictive performance. The results underscore the importance of a robust preprocessing pipeline in Raman spectroscopy analysis and suggest that retaining the full set of preprocessed features yields more accurate and reliable classification outcomes.

References

1. Suriyah Akhter, Mohd Aadil Bhat, Sirajuddin Ahmed, and Weqar Ahmed Siddiqui. Antibiotic residue contamination in the aquatic environment, sources and associated potential health risks. *Environmental Geochemistry and Health* 2024 46:10, 46:1–27, 8 2024.
2. Sarah J. Pitt and Alan Gunn. The one health concept. *British Journal of Biomedical Science*, 81:12366, 2 2024.
3. Alice Caneschi, Anisa Bardhi, Andrea Barbarossa, and Anna Zaghini. The use of antibiotics and antimicrobial resistance in veterinary medicine, a complex phenomenon: A narrative review. *Antibiotics* 2023, Vol. 12, Page 487, 12:487, 3 2023.
4. Felipe C. Cabello, Henry P. Godfrey, Alejandro H. Buschmann, and Humberto J. Dölz. Aquaculture as yet another environmental gateway to the development and globalisation of antimicrobial resistance. *The Lancet Infectious Diseases*, 16:e127–e133, 7 2016.
5. Ivo Schmerold, Inge van Geijlswijk, and Ronette Gehring. European regulations on the use of antibiotics in veterinary medicine. *European Journal of Pharmaceutical Sciences*, 189:106473, 10 2023.
6. Pia Väitalo, Antonina Kruglova, Anna Mikola, and Riku Vahala. Toxicological impacts of antibiotics on aquatic micro-organisms: A mini-review. *International Journal of Hygiene and Environmental Health*, 220:558–569, 5 2017.
7. Joachim K. Seydel. Sulfonamides, structure-activity relationship, and mode of action: Structural problems of the antibacterial action of 4-aminobenzoic acid (paba) antagonists. *Journal of Pharmaceutical Sciences*, 57:1455–1478, 9 1968.
8. Aben Ovung and Jhimli Bhattacharyya. Sulfonamide drugs: structure, antibacterial property, toxicity, and biophysical interactions. *Biophysical Reviews* 2021 13:2, 13:259–272, 3 2021.
9. Zeinab Maghsodian, Ali Mohammad Sanati, Tebogo Mashifana, Mika Sillanpää, Shengyu Feng, Tan Nhat, and Bahman Ramavandi. Occurrence and distribution

- of antibiotics in the water, sediment, and biota of freshwater and marine environments: A review. *Antibiotics* 2022, Vol. 11, Page 1461, 11:1461, 10 2022.
10. World Health Organization. Critically important antimicrobials for human medicine 6th re.vision 2018. ranking of medically important antimicrobials for risk management of antimicrobial resistance due to non-human use., 2018.
 11. D. Fatta, A. Achilleos, A. Nikolaou, and S. Meriç. Analytical methods for tracing pharmaceutical residues in water and wastewater. *TrAC Trends in Analytical Chemistry*, 26:515–533, 6 2007.
 12. Zhi Min Zhang, Shan Chen, Yi Zeng Liang, Zhao Xia Liu, Qi Ming Zhang, Li Xia Ding, Fei Ye, and Hua Zhou. An intelligent background-correction algorithm for highly fluorescent samples in raman spectroscopy. *Journal of Raman Spectroscopy*, 41:659–669, 6 2010.
 13. Qi Wang and Wan Ming Zhao. Optical methods of antibiotic residues detections: A comprehensive review. *Sensors and Actuators B: Chemical*, 269:238–256, 9 2018.
 14. Jisue Moon, Meijun Li, Anibal J. Ramirez-Cuesta, and Zili Wu. Raman spectroscopy. pages 75–110. Springer, Cham, 2023.
 15. Yazhou Qin, Jiaxin Qiu, Nan Tang, Yingsheng He, and Li Fan. Deep learning analysis for rapid detection and classification of household plastics based on raman spectroscopy. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 309:123854, 3 2024.
 16. Oleg Ryabchykov, Juergen Popp, and Thomas Bocklitz. Fusion of maldi spectrometric imaging and raman spectroscopic data for the analysis of biological samples. *Frontiers in Chemistry*, 6:341945, 7 2018.
 17. Paul R. Carey and Hossein Heidari-Torkabadi. New techniques in antibiotic discovery and resistance: Raman spectroscopy. *Annals of the New York Academy of Sciences*, 1354:67–81, 9 2015.
 18. Karin Wårdell, Kerstin Ramser, Martin Hallbeck, Jan Hillman, Joel Wahl, and Elisabeth Klint. Impact of preprocessing methods on the raman spectra of brain tissue. *Biomedical Optics Express*, Vol. 13, Issue 12, pp. 6763–6777, 13:6763–6777, 12 2022.
 19. Kenny Kong, Catherine Kendall, Nicholas Stone, and Ioan Notingher. Raman spectroscopy for medical diagnostics — from in-vitro biofluid assays to in-vivo cancer detection. *Advanced Drug Delivery Reviews*, 89:121–134, 7 2015.
 20. Gurvinder Singh Bumrah and Rakesh Mohan Sharma. Raman spectroscopy – basic principle, instrumentation and selected applications for the characterization of drugs of abuse. *Egyptian Journal of Forensic Sciences*, 6:209–215, 9 2016.
 21. Nikita Lakomkin and Constantinos G. Hadjipanayis. The use of spectroscopy handheld tools in brain tumor surgery: Current evidence and techniques. *Frontiers in Surgery*, 6:449338, 5 2019.
 22. Shiyang Fang, Siyi Wu, Zhou Chen, Chang He, Linley Li Lin, and Jian Ye. Recent progress and applications of raman spectrum denoising algorithms in chemical and biological analyses: A review. *TrAC Trends in Analytical Chemistry*, 172:117578, 3 2024.
 23. K. Chen, Y. H. Ong, C. Yuen, and Q. Liu. Surface-enhanced raman spectroscopy for intradermal measurements. pages 141–154. Elsevier Inc., 8 2016.
 24. Anna Lipovka, Maxim Fatkullin, Andrey Averkiev, Marina Pavlova, Anurag Adiraju, Saddam Weheabby, Ammar Al-Hamry, Olfa Kanoun, Igor Pašti, Tamara Lazarevic-Pasti, Raul D. Rodriguez, and Evgeniya Sheremet. Surface-enhanced raman spectroscopy and electrochemistry: The ultimate chemical sensing and manipulation combination. *Critical Reviews in Analytical Chemistry*, 54:110–134, 2024.

25. Yaping Qi, Dan Hu, Yucheng Jiang, Zhenping Wu, Ming Zheng, Esther Xinyi Chen, Yong Liang, Mohammad A. Sadi, Kang Zhang, and Yong P. Chen. Recent progresses in machine learning assisted raman spectroscopy. *Advanced Optical Materials*, 11:2203104, 7 2023.
26. İbrahim Yazici, İbraheem Shayea, and Jafri Din. A survey of applications of artificial intelligence and machine learning in future mobile networks-enabled systems. *Engineering Science and Technology, an International Journal*, 44, 8 2023.
27. Eleftherios Pavlou and Nikolaos Kourkouvelis. Preprocessing and analyzing raman spectra using python. *Engineering Proceedings 2023, Vol. 56, Page 28*, 56:28, 10 2023.
28. Thomas Bocklitz, Angela Walter, Katharina Hartmann, Petra Rösch, and Jürgen Popp. How to pre-process raman spectra for reliable and stable models? *Analytica Chimica Acta*, 704:47–56, 10 2011.
29. Rekha Gautam, Sandeep Vanga, Freek Ariese, and Siva Umapathy. Review of multidimensional data processing approaches for raman and infrared spectroscopy. *EPJ Techniques and Instrumentation 2015 2:1*, 2:1–38, 6 2015.
30. Alsamad Fatima, Gobinet Cyril, Vuiblet Vincent, Jaisson Stéphane, and Piot Olivier. Towards normalization selection of raman data in the context of protein glycation: application of validity indices to pca processed spectra. *Analyst*, 145:2945–2957, 4 2020.
31. P Lasch and Peter Lasch. Spectral pre-processing for biomedical vibrational spectroscopy and microspectroscopic imaging spectral pre-processing for vibrational spectroscopy spectral pre-processing for biomedical vibrational spectroscopy and microspectroscopic imaging. 117:100–114, 2012.
32. Walker Peterson, Julia Gala De Pablo, Matthew Lindley, Kotaro Hiramatsu, and Keisuke Goda. Ultrafast impulsive raman spectroscopy across the terahertz-fingerprint region.
33. Nicolas Coca-Lopez. An intuitive approach for spike removal in raman spectra based on peaks’ prominence and width. *Analytica Chimica Acta*, 1295, 3 2024.
34. Darren Whitaker, Kevin Hayes, and Darren A Whitaker. A simple algorithm for despiking raman spectra.
35. H. Georg Schulze and Robin F.B. Turnera. A two-dimensionally coincident second difference cosmic ray spike removal method for the fully automated processing of raman spectra. *Applied Spectroscopy*, 68:185–191, 2 2014.
36. Xihui Bian, Zitong Shi, Yingjie Shao, Yuanyuan Chu, and Xiaoyao Tan. Variational mode decomposition for raman spectral denoising. *Molecules 2023, Vol. 28, Page 6406*, 28:6406, 9 2023.
37. Abraham Savitzky and Marcel J.E. Golay. Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, 36:1627–1639, 7 1964.
38. Hao Chen, Weiliang Xu, Neil Broderick, and Jianda Han. An adaptive denoising method for raman spectroscopy based on lifting wavelet transform. *Journal of Raman Spectroscopy*, 49:1529–1539, 9 2018.
39. Yixin Guo, Weiqi Jin, Zongyu Guo, and Yuqing He. Iterative differential autoregressive spectrum estimation for raman spectrum denoising. *Journal of Raman Spectroscopy*, 53:148–165, 1 2022.
40. Xinyuan Bi, Li Lin, Zhou Chen, and Jian Ye. Artificial intelligence for surface-enhanced raman spectroscopy. *Small Methods*, 8, 1 2024.
41. Alexis Lebrun, Hubert Fortin, Nicolas Fontaine, Daniel Fillion, Olivier Barbier, and Denis Boudreau. Pushing the limits of surface-enhanced raman spectroscopy (sers) with deep learning: Identification of multiple species with closely related molecular structures. *Applied Spectroscopy*, 76:609, 5 2022.

42. Yunnan Xu, Pang Du, Ryan Senger, John Robertson, and James L Pirkle. Isrea: An efficient peak-preserving baseline correction algorithm for raman spectra.
43. M ; Han, Y ; Dang, J Han, Ming Han, Yu Dang, and Jianda Han. Denoising and baseline correction methods for raman spectroscopy based on convolutional autoencoder: A unified solution. *Sensors (Basel, Switzerland)*, 24:3161, 5 2024.
44. M. Hamed Mozaffari and Li-Lin Tay. One-dimensional active contour models for raman spectrum baseline correction, 4 2021.
45. A. P. Dempster. An overview of multivariate data analysis. *Journal of Multivariate Analysis*, 1:316–346, 9 1971.
46. Alex Diaz-Papkovich, Luke Anderson-Trocmé, Chief Ben-Eghan, and Simon Gravel. Umap reveals cryptic population structure and phenotype heterogeneity in large genomic cohorts. *PLOS Genetics*, 15:e1008432, 2019.
47. Martin Wattenberg, Fernanda Viégas, and Ian Johnson. How to use t-sne effectively. *Distill*, 1:e2, 10 2016.
48. Leland McInnes, John Healy, and James Melville. Umap: Uniform manifold approximation and projection for dimension reduction, 2020.
49. Cecil C. Bridges. Hierarchical cluster analysis. *Psychological Reports*, 18:851–854, 6 1966.
50. Slamet Widodo, Herlambang Brawijaya, and Samudi Samudi. Stratified k-fold cross validation optimization on machine learning for prediction. *Sinkron : jurnal dan penelitian teknik informatika*, 6:2407–2414, 10 2022.
51. Effendi Widjaja, Wei Zheng, and Zhiwei Huang. Classification of colonic tissues using near-infrared raman spectroscopy and support vector machines. *International Journal of Oncology*, 32:653–662, 3 2008.
52. Patrick Buzzini, James Curran, and Carrie Polston. Comparison between visual assessments and different variants of linear discriminant analysis to the classification of raman patterns of inkjet printer inks. *Forensic Chemistry*, 24:100336, 6 2021.
53. Petros Xanthopoulos, Panos M. Pardalos, and Theodore B. Trafalis. Linear discriminant analysis. pages 27–33. Springer, New York, NY, 2013.
54. Fa Zhu, Junbin Gao, Jian Yang, and Ning Ye. Neighborhood linear discriminant analysis. *Pattern Recognition*, 123:108422, 3 2022.
55. Shiyang Fang, Siyi Wu, Zhou Chen, Chang He, Linley Li Lin, and Jian Ye. Recent progress and applications of raman spectrum denoising algorithms in chemical and biological analyses: A review. *TrAC Trends in Analytical Chemistry*, 172:117578, 3 2024.
56. Karin Wårdell, Kerstin Ramser, Martin Hallbeck, Jan Hillman, Joel Wahl, and Elisabeth Klint. Impact of preprocessing methods on the raman spectra of brain tissue. *Biomedical Optics Express*, Vol. 13, Issue 12, pp. 6763-6777, 13:6763–6777, 12 2022.
57. Shiyang Fang, Siyi Wu, Zhou Chen, Chang He, Linley Li Lin, and Jian Ye. Recent progress and applications of raman spectrum denoising algorithms in chemical and biological analyses: A review. *TrAC Trends in Analytical Chemistry*, 172:117578, 3 2024.
58. K. Chen, Y. H. Ong, C. Yuen, and Q. Liu. Surface-enhanced raman spectroscopy for intradermal measurements. *Imaging in Dermatology*, pages 141–154, 8 2016.